

Exposé on Master's Thesis

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Title: Stereo Camera-Based Environment Perception for Automated Vehicles
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1 Topic

Autonomous vehicles are rapidly becoming a widespread reality. However, their perception systems still face significant challenges in complex and unforeseen scenarios, often leading to disengagements, where human intervention via teleoperation is required. In such critical moments, the vehicle must rely on an accurate and robust perception system to ensure safe navigation.

Most existing perception systems, responsible for detecting and tracking objects in the vehicle's surroundings, rely heavily on learning-based solutions. While these methods have demonstrated impressive performance in structured and controlled environments, they often struggle to generalize in novel and untrained scenarios. This limitation is particularly problematic in safety-critical situations, where the vehicle must make quick decisions based on the perceived environment.

This project aims to develop a deterministic, stereo camera-based perception system for autonomous vehicles, capable of providing a reliable and robust understanding of the vehicle's surroundings, even in highly dynamic and unforeseen scenarios. The system will take as input two images from a stereo camera mounted on the vehicle and output a list of detected objects in the scene, along with their criticality scores, forming a situational awareness module to assess potential hazards. Additionally, this deterministic approach could run in parallel with existing learning-based perception systems, providing an additional layer of redundancy and security in critical situations where the learning-based methods may fail.

To achieve accurate 6D motion estimation (position + velocity), the system will implement a deterministic algorithm inspired by the Daimler 6D Vision system. This approach uses Kalman Filters to estimate a 6D state vector for selected feature points in the scene, which will then be clustered to identify moving objects. Unlike learning-based methods, this approach does not require extensive training data and can generalize well in new unseen scenarios.

Given the real-time requirements of autonomous vehicles, the system will be implemented in C++ using the Robot Operating System (ROS) framework. It will also incorporate GPU hardware acceleration using CUDA to speed up computationally intensive tasks, such as feature point detection, clustering or optical flow computation.

To ensure scalability and modularity, the entire system will be containerized using Docker, facilitating deployment across different hardware configurations. Finally, the developed system will be deployed and evaluated on the EDGAR research vehicle. Here, the system's performance will be compared against the state-of-the-art learning-based perception systems currently used in the vehicle.

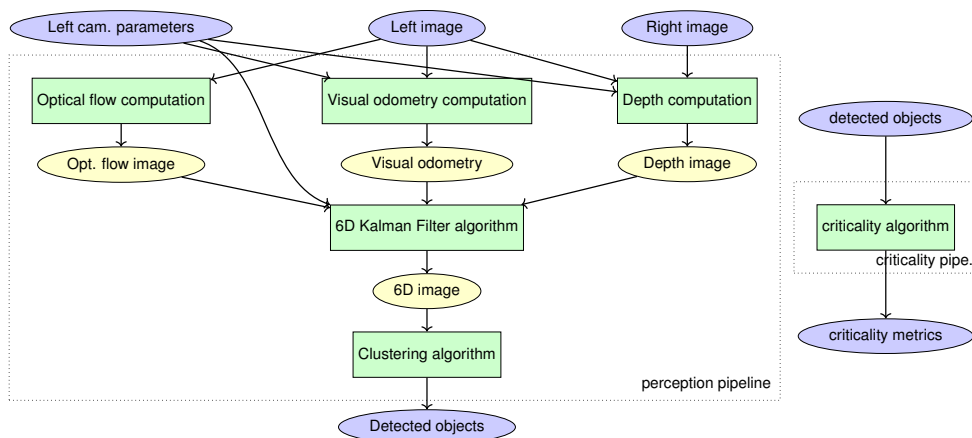


Figure 1: Side-by-side diagrams of the high level algorithms

2 State of Science and Technology

Things to talk about:

1. Perception systems in autonomous vehicles, why they are important, what they do, how they work, what are the challenges, what are the current solutions, what are the limitations, what are the future trends. 2. Type of sensors used for perception in autonomous vehicles 3. What 6D vision is and how it works

Lets do some tests,

General survey about autonomous vehicles perception: [[Van+18](#)]

The Daimler vision stuff [[Fra+05](#)]

3 Objective

Citation: [choset2005principles; thrun2005probabilistic]

Reference: fig. 2

Figures:

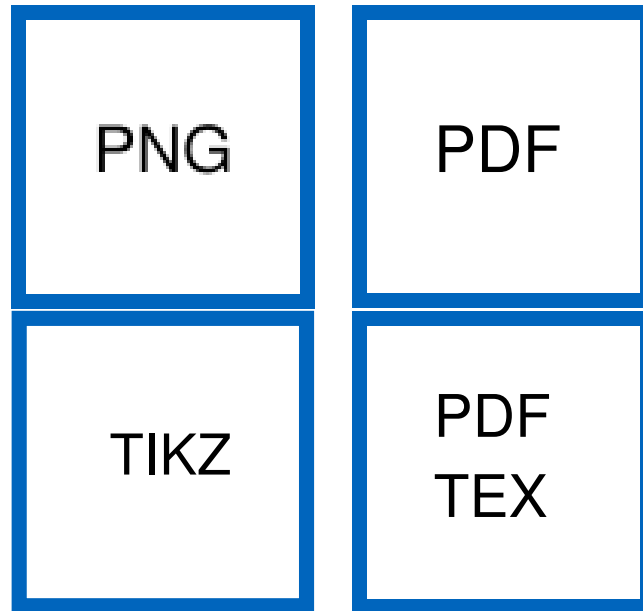


Figure 2: Image description.

4 Work Plan and necessary Resources

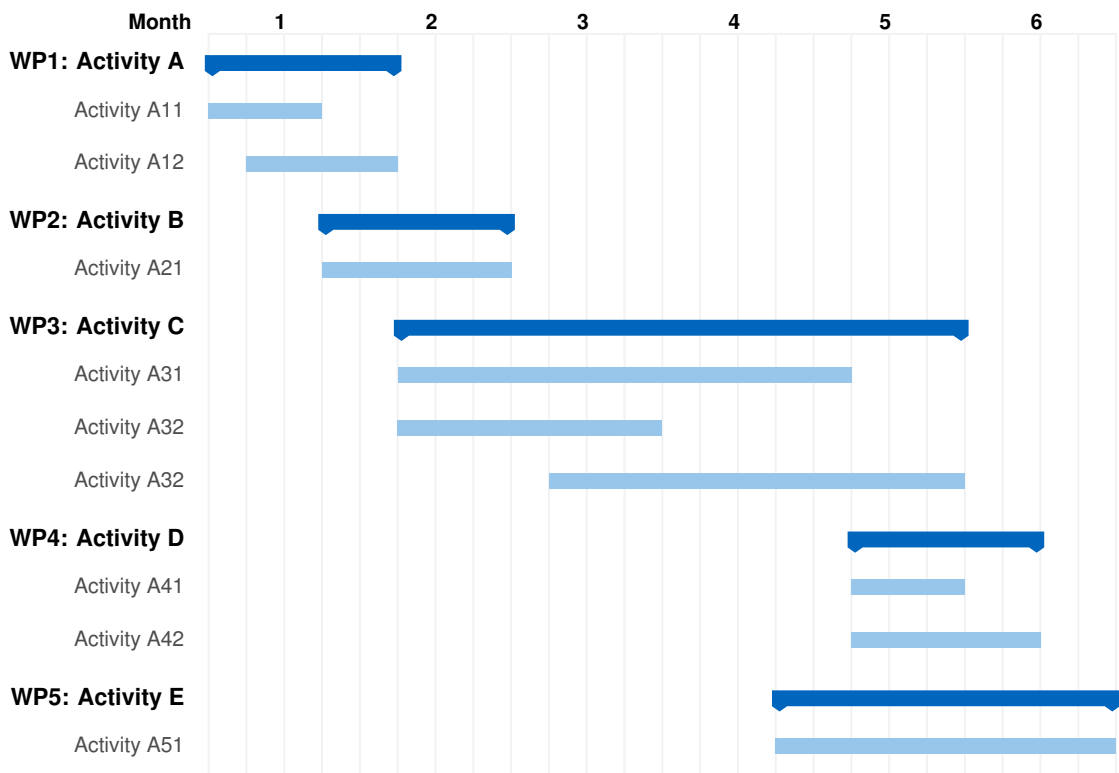


Figure 3: Time schedule.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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5 Literature

- [Fra+05] Franke, U., Rabe, C., Badino, H., and Gehrig, S. “6D-Vision: Fusion of Stereo and Motion for Robust Environment Perception”. en. In: *Pattern Recognition*. Ed. by Kropatsch, W. G., Sablatnig, R., and Hanbury, A. Berlin, Heidelberg: Springer, 2005, pp. 216–223. ISBN: 978-3-540-31942-9. DOI: [10.1007/11550518_27](https://doi.org/10.1007/11550518_27).
- [Van+18] Van Brummelen, J., O’Brien, M., Gruyer, D., and Najjaran, H. “Autonomous vehicle perception: The technology of today and tomorrow”. en. In: *Transportation Research Part C: Emerging Technologies* 89 (Apr. 2018), pp. 384–406. ISSN: 0968090X. DOI: [10.1016/j.trc.2018.02.012](https://doi.org/10.1016/j.trc.2018.02.012). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0968090X18302134> (visited on 10/25/2024).